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Sustainable Synthesis of NiO Nanoflowers: a Low-Cost Hydrothermal Strategy without Autoclave

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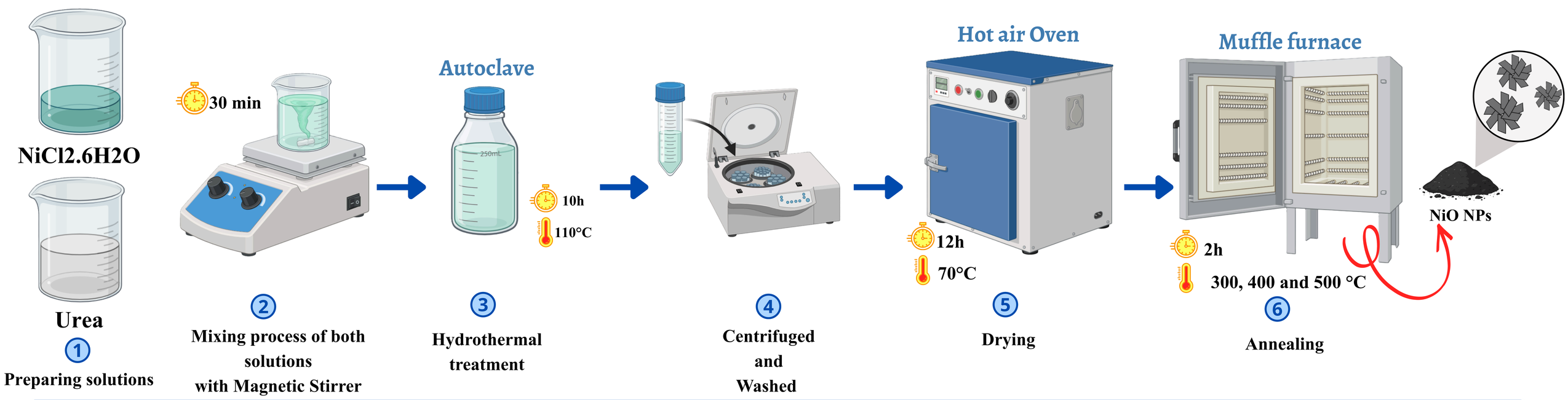
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Introduction :

Nanoflowers offer significant advantages in supercapacitors due to their unique, hierarchical, and flower-like structure, which provides a large specific surface area. They enhance electrochemical performance by providing abundant active sites, improving ion diffusion and electron transport, and increasing structural stability during cycling, typically leading to high specific capacitance, long-term cyclability, and superior power density [1]. This Nanoflowers can be synthesized using various methods such as Physical vapor deposition, hydrothermal , Green synthesis , and Electrochemical deposition, each offering distinct advantages and influencing the properties of the nanostructures differently. Thus, optimizing synthesis methods to produce flower-like NiO with tailored properties remains a critical challenge in nanotechnology.This work demonstrates a novel hydrothermal approach using ordinary laboratory glassware instead of conventional autoclaves, combined with green chemistry principles, to produce high-quality NiO nanoflowers without compromising efficiency.

1-Synthesis of NiO Flower-Like structures by Hydrothermal Method :



2-Results and Discussion :

XRD Analysis :

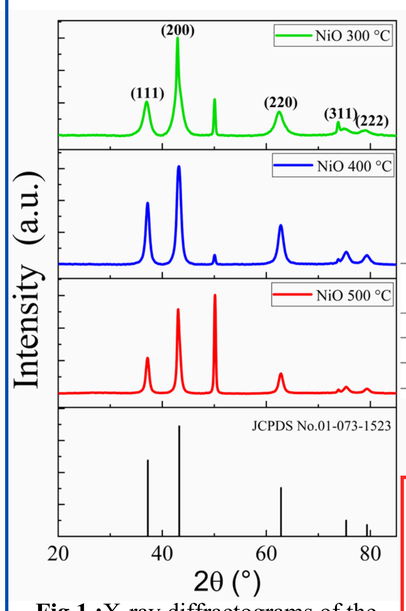


Fig 1 :X-ray diffractograms of the specimens annealed at different temperatures (300 °C, 400 °C and 500 °C) [2]

- Peak indexing was performed using X'PertHighScore© software.
- Prepared NiO exhibited a face-centered cubic (FCC) structure .

Table 1 : W–H extracted parameters

Samples	Crystallite size D (nm)	Microstrain ε	Dislocation densities $\delta = 1/D^2$ (nm^{-2})
NiO–300 °C	22.45	0.006	0.0020
NiO–400 °C	31.66	0.005	0.0010
NiO–500 °C	43.33	0.004	0.0005

The decreasing microstrain and dislocation density with temperature confirm progressive relaxation of internal stresses and improvement in crystalline quality.

RAMAN Spectra:

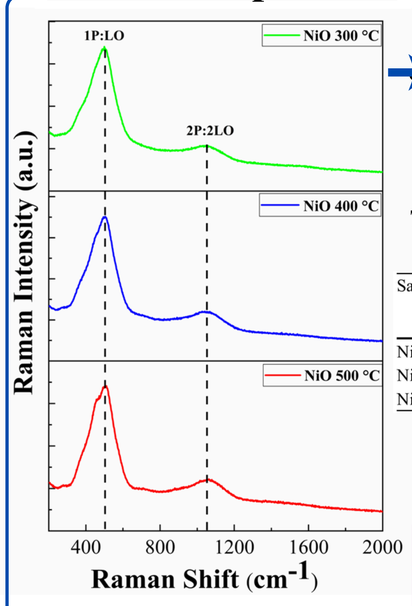


Fig 2 : Raman spectra of NiO NPs annealed at different temperatures (300 °C, 400 °C, and 500 °C) [2]

Exhibit two main features the one-phonon (1LO) and two-phonon (2LO) modes of cubic NiO (Fm-3m)

Table 2 : Raman peak positions and FWHM (Lorentzian fits)

Samples	1LO position (cm^{-1})	1LO FWHM (cm^{-1})	2LO position (cm^{-1})	2LO FWHM (cm^{-1})
NiO–300 °C	477.00	231.43	1038.73	392.62
NiO–400 °C	480.77	212.80	1039.31	374.95
NiO–500 °C	484.09	180.13	1049.00	405.60

- Progressive relaxation of lattice strain .
- Reduction in structural disorder .
- Improve crystallinity and decreased phonon scattering .

SEM Analysis :

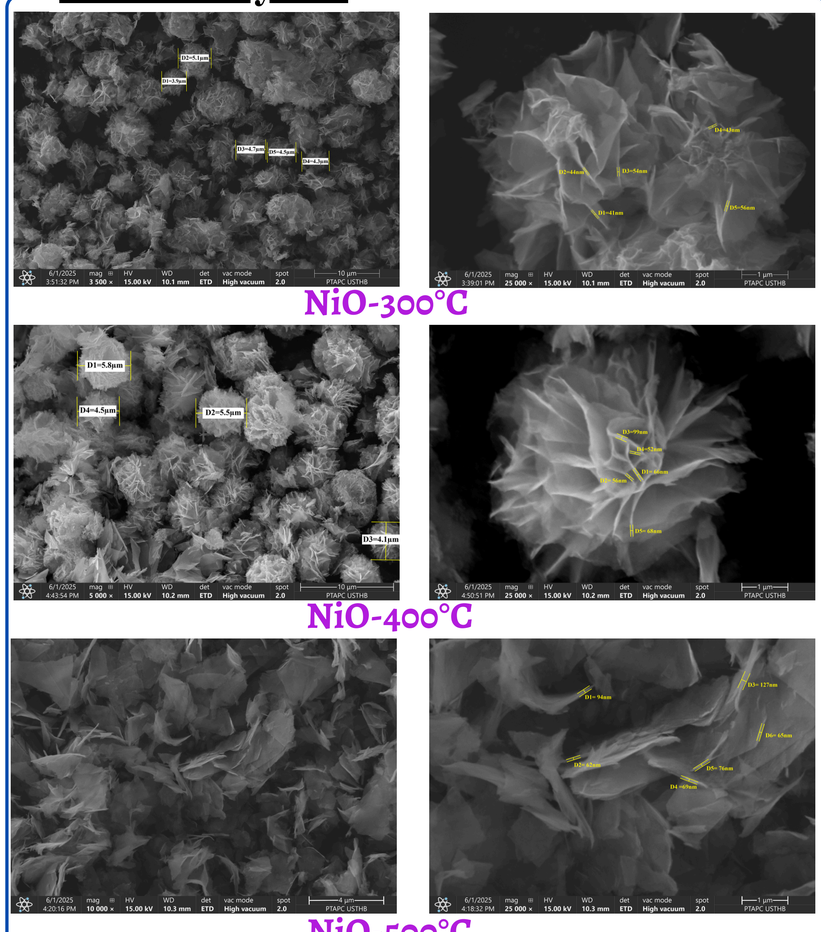


Fig 4 : SEM micrographs of nanoflower-like NiO annealed at different temperatures[2]

VSM Analysis :

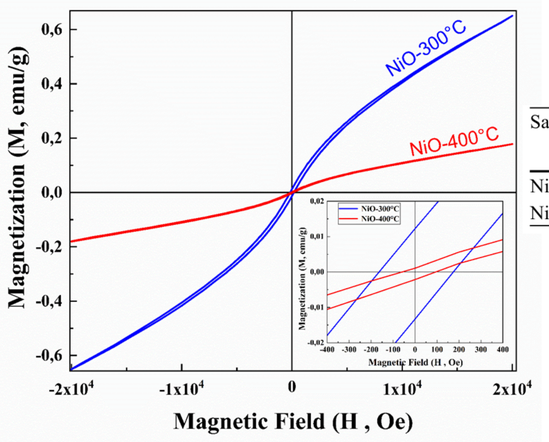


Fig 3: Room temperature obtained Magnetic (M-H) curves of NiO Flower-Like structure annealed at 300 °C and 400 °C [2]

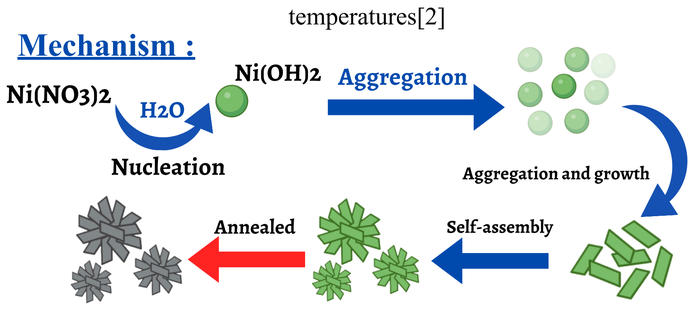
Table 2 : Magnetic parameters of NiO Nanoparticles synthesized by hydrothermal

Samples	Annealing temperature (°C)	VSM of Hysteresis parameters $H_c(Oe)$	$M_m \times 10^{-3}(emu/g)$	$M_r \times 10^{-3}(emu/g)$	Loop area (Oe × emu/g)	SQR = M_r/M_s SQR
NiO-300 °C	300 °C	169.04	650.92	12.77	400.62	0.02
NiO-400 °C	400 °C	76.31	179.49	1.60	92.41	8.91×10^{-3}

Weak ferromagnetic behavior

progressive reduction of magnetic anisotropy and a transition toward softer magnetic behavior.

Mechanism :



3-Conclusion & Future works:

- Successful synthesis of NiO Flower-Like structures with notable characteristics using a a green and low-cost hydrothermal process that preferably uses borosilicate glass bottles instead of conventional Teflon-lined autoclaves.
- The annealing temperature is a key tuning factor in the development of NiO nanoflowers for specific applications based on clear evidence of structure-morphology correlation and magnetic properties.
- Future research may also focus on:
  - Surface functionalization towards higher selectivity and sensitivity in sensors.
  - Electrochemical assessment for supercapacitor and battery electrodes .
  - Catalytic assessment of pollutant degradation in waste water treatment.

4-References :

[1]Su Jung, L., Hongje, J., Do Nam, L.: Recent advances in nanoflowers: compositional and structural diversification for potential applications. Nanoscale Adv. (2023). https://doi.org/10.1039/d3na00163f

[2]Derbil, S., Mouaci, S., Benchikh Lehoucine, C., Rabia, S., Younes, A., Ait-Kaci, A., Saidi, M ., Saidi-Amroun, N. : Sustainable Synthesis of NiO Nanoflowers: a Low-Cost Hydrothermal Strategy without Autoclave. J Supercond Nov Magn (2025). https://doi.org/10.1007/s10948-025-07092-6